

Linear Relations

Marking Scheme & Blueprint

MTH1W · Module 5 Test — Teacher Copy

Blueprint

Curriculum: MTH1W (Grade 9 de-streamed math). Total: 70 marks · 60 minutes · 30 questions. Categories: K/U, Application (App), Thinking (Think), Communication (Comm).

Q	Sub-topic	Level	Marks	Category	Section
A1	First Differences	L1	1	K/U	A
A2	Variation	L1	1	K/U	A
A3	Variation	L1	1	K/U	A
A4	Slope	L1	1	K/U	A
A5	Intercepts	L1	1	K/U	A
A6	Slope	L1	1	K/U	A
A7	Intercepts	L1	1	K/U	A
A8	Standard Form	L1	1	K/U	A
A9	Correlation	L1	1	K/U	A
A10	First Differences	L1	2	K/U	A
A11	Slope	L1	2	K/U	A
A12	Intercepts	L1	2	K/U	A
A13	Equation of a Line	L1	2	K/U	A
A14	Variation	L1	2	K/U	A
A15	Standard Form	L1	2	K/U	A
B16	Slope	L2	2	App	B
B17	Intercepts	L2	2	App	B
B18	Equation of a Line	L2	2	App	B
B19	First Differences	L2	2	App	B
B20	Standard Form	L2	2	App	B
B21	Variation	L2	2	App	B
B22	Line Through Two Points	L2	3	App	B
B23	Equation from Intercepts	L2	3	App	B
B24	Linear Model	L2	3	App	B
C25	Unknown Coordinate	L3	4	Think	C
C26	Missing Table Value	L3	4	Think	C
C27	Linear Model	L3	5	Think	C
C28	Slope, Equation & Intercept	L3	5	Think	C
C29	Point of Intersection	L3	6	Think	C
D30	Variation vs Linearity	L3	4	Comm	D

By category (marks): K/U 21 · Application 21 · Thinking 24 · Communication 4

By level (marks): L1 21 · L2 21 · L3 28 → L1 30% · L2 30% · L3 40%

Worked Solutions & Mark Allocation

- 1. (A1 · K/U · 1 marks)**
4, 4, 4. (A1)
- 2. (A2 · K/U · 1 marks)**
Direct. (A1)
- 3. (A3 · K/U · 1 marks)**
Partial. (A1)
- 4. (A4 · K/U · 1 marks)**
 $12/3 = 4$. (A1)
- 5. (A5 · K/U · 1 marks)**
9, at (0, 9). (A1)
- 6. (A6 · K/U · 1 marks)**
-3. (A1)
- 7. (A7 · K/U · 1 marks)**
 $0 = x - 4 \rightarrow x = 4$, at (4, 0). (A1)
- 8. (A8 · K/U · 1 marks)**
 $5x - y + 2 = 0$. (A1)
- 9. (A9 · K/U · 1 marks)**
Positive. (A1)
- 10. (A10 · K/U · 2 marks)**
6, 6, 6; constant, so linear. (M1, A1)
- 11. (A11 · K/U · 2 marks)**
 $(14-2)/(4-1) = 12/3 = 4$. (M1, A1)
- 12. (A12 · K/U · 2 marks)**
y-int (0, -8); x-int: $0 = 2x - 8 \rightarrow (4, 0)$. (M1, A1)
- 13. (A13 · K/U · 2 marks)**
 $y = 3x - 5$. (M1, A1)
- 14. (A14 · K/U · 2 marks)**
 $y = 4x + 10$. (M1, A1)
- 15. (A15 · K/U · 2 marks)**
Multiply by 4: $4y = 3x + 8$, so $3x - 4y + 8 = 0$. (M1, A1)
- 16. (B16 · Application · 2 marks)**
 $\text{slope} = (-1 - 7)/(2 - (-2)) = -8/4 = -2$. (M1, A1)

17. (B17 • Application • 2 marks)

x-int: $y = 0 \rightarrow 2x = 20 \rightarrow (10, 0)$. (M1)

y-int: $x = 0 \rightarrow 5y = 20 \rightarrow (0, 4)$. (A1)

18. (B18 • Application • 2 marks)

$7 = 5(2) + b \rightarrow b = -3$. (M1)

$y = 5x - 3$. (A1)

19. (B19 • Application • 2 marks)

Differences 5, 7, 9 are not constant. (M1)

So the relation is non-linear. (A1)

20. (B20 • Application • 2 marks)

Multiply by 4: $4y = 3x - 8$. (M1)

$3x - 4y - 8 = 0$. (A1)

21. (B21 • Application • 2 marks)

$k = 24/6 = 4$. (M1)

$y = 4x$. (A1)

22. (B22 • Application • 3 marks)

slope = $(16-4)/(5-1) = 12/4 = 3$. (M1)

$4 = 3(1) + b \rightarrow b = 1$. (M1)

$y = 3x + 1$. (A1)

23. (B23 • Application • 3 marks)

Through (3, 0) and (0, -6): slope = $(-6-0)/(0-3) = 2$. (M1, M1)

$y = 2x - 6$. (A1)

24. (B24 • Application • 3 marks)

$V = 12(10) + 30 = 120 + 30 = 150$ L. (M1, A1)

Initial value is 30 L. (A1)

25. (C25 • Thinking • 4 marks)

$(21 - 9)/(k - 2) = 4$. (M1)

$12 = 4(k - 2) \rightarrow k - 2 = 3$. (M1)

$k = 5$. (A1, A1)

26. (C26 • Thinking • 4 marks)

Linear $\rightarrow$ equal first differences; from -5 to 7 over two steps is +12, so +6 each. (M1)

Missing value: $-5 + 6 = 1$. (A1)

y-intercept -5, slope 6 $\rightarrow y = 6x - 5$. (M1, A1)

27. (C27 • Thinking • 5 marks)

$C = 0.20m + 30$. (M1)

At $m = 150$: $C = 0.20(150) + 30 = 30 + 30 = \60 . (M1, A1)

For \$50: $0.20m + 30 = 50 \rightarrow 0.20m = 20 \rightarrow m = 100$ minutes. (M1, A1)

28. (C28 • Thinking • 5 marks)

$$\text{slope} = (11 - (-1))/(3 - (-3)) = 12/6 = 2. \text{ (M1)}$$

$$-1 = 2(-3) + b \rightarrow b = 5, \text{ so } y = 2x + 5. \text{ (M1, A1)}$$

$$\text{x-int: } 0 = 2x + 5 \rightarrow x = -5/2, \text{ at } (-5/2, 0). \text{ (M1, A1)}$$

29. (C29 • Thinking • 6 marks)

$$\text{Line B slope} = (4 - 7)/(4 - 1) = -3/3 = -1. \text{ (M1)}$$

$$7 = -1(1) + b \rightarrow b = 8, \text{ so } y = -x + 8. \text{ (M1, A1)}$$

$$\text{Intersect: } 2x - 1 = -x + 8 \rightarrow 3x = 9 \rightarrow x = 3. \text{ (M1)}$$

$$y = 2(3) - 1 = 5, \text{ so } (3, 5). \text{ (A1, A1)}$$

30. (D30 • Communication • 4 marks)

Direct variation must pass through the origin, i.e. have y-intercept 0. (C1)

Counter-example: $y = 2x + 3$ is a straight line but its y-intercept is $3 \neq 0$, so it is partial, not direct. (A1, C1)

A linear relation is a direct variation only when its y-intercept is 0 (form $y = kx$). (C1)